

# README



## Боба / Boba

**Multi-tracker meta-search for qBittorrent — self-hosted, containerised, private-tracker-aware.**

[Quick start](#) · [Tokens & keys](#) · [Documentation](#) · [Architecture](#) · [Testing](#) · [Contributing](#)

|               |                |                |                                                                       |                      |    |
|---------------|----------------|----------------|-----------------------------------------------------------------------|----------------------|----|
| python tests  | 6271 collected | frontend tests | 2743 collected                                                        | plugins              | 43 |
| merge service | FastAPI :7187  | ci             | manual (./ci.sh, no auto-trigger)                                     | pre-build invariants | 58 |
| challenges    | 38             | scanners       | snyk   sonar   bandit   ruff   semgrep   trivy   gitleaks   pip-audit |                      |    |
| license       | Apache 2.0     |                |                                                                       |                      |    |

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## Features

- **Merge Search Service** — FastAPI service (:7187) that fans out across 40+ trackers, deduplicates results, streams via SSE.
- **Real-time results** — `result_found` events arrive as each tracker completes, no blocking.

- **Private-tracker bridge** — Authenticated downloads via boba-webui-bridge.service (the Go webui-bridge binary, :7188) for RuTracker, Kinozal, NNM-Club, IPTorrents. The original webui-bridge.py implementation is kept in the tree but is not part of the managed boba.target topology.
  - **Freeleech-only IPTorrents** — Automation never costs ratio (see [constitution VIII](#)).
  - **Opt-in quality stack** — SonarQube + Snyk + Semgrep + Trivy + Gitleaks + bandit + pip-audit behind docker-compose.quality.yml.
  - **Opt-in observability** — Prometheus + Grafana dashboards behind the same profile system.
  - **Hook system** — Register scripts via POST /api/v1/hooks for search / download events.
  - **Angular 21 SPA dashboard** — dark-themed, signals-based, per-tracker status chips with CAPTCHA re-login, virtual-scroll-ready sort.
  - **PWA-ready** — favicon + launcher icons + web manifest ship from frontend/public/.
  - **Container-first** — rootless Podman or Docker, single ./start.sh boot.
  - **Resource-pressure early-warning** — [challenges/scripts/resource\\_pressure\\_signature\\_challenge.sh](#) checks 5 independently-falsifiable host-pressure signatures (runaway RSS, thread-count vs ulimit -u, container EAGAIN cascades, swap pressure, cgroup memory pressure) so a forced-logout precursor is caught *before* user@1000.service gets SIGKILLed, not after. Wired into scripts/pre\_build\_verification.sh invariant 25 and a standing boba-resource-pressure-check.timer systemd-user unit. See [incident writeup](#).
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## Quick start

```
git clone https://github.com/milos85vasic/qBitTorrent.git
cd qBitTorrent
```

*# Optional: configure tracker credentials + tokens*

```
cp .env.example .env
```

```
$EDITOR .env # see "Tokens & API keys" below
```

```
./setup.sh # one-time
```

```
./start.sh # Angular build + containers + plugin sync
```

Then visit:

| URL                                                                                 | Purpose                                  | Auth          |
|-------------------------------------------------------------------------------------|------------------------------------------|---------------|
| <a href="http://localhost:7187/">http://localhost:7187/</a>                         | Merge-search dashboard (SPA)             | —             |
| <a href="http://localhost:7187/docs">http://localhost:7187/docs</a>                 | FastAPI Swagger UI                       | —             |
| <a href="http://localhost:7187/openapi.json">http://localhost:7187/openapi.json</a> | OpenAPI schema                           | —             |
| <a href="http://localhost:7186/">http://localhost:7186/</a>                         | qBittorrent WebUI proxy                  | admin / admin |
| <a href="http://localhost:7185/">http://localhost:7185/</a>                         | qBittorrent internal (container)         | —             |
| <a href="http://localhost:7188/">http://localhost:7188/</a>                         | WebUI bridge (private-tracker downloads) | —             |

## Tokens & API keys

⚠ **Configure any credentials you need in `.env` before running `./start.sh`.** The single source of truth is [docs/TOKENS\\_AND\\_KEYS.md](#) — which variable is mandatory, which is optional, and **where to register** for each one.

## Tracker credentials — env vars AND cookies files (auto-loaded)

Boba supports **two mechanisms** for tracker auth, and both are respected together:

1. **Env vars in `.env`** — `<TRACKER>_USERNAME / <TRACKER>_PASSWORD / <TRACKER>_COOKIES`.
2. **Cookies files in `~/Downloads/`** — canonical naming `cookies_<tracker>.txt` (lowercase), Netscape TSV format exported from a logged-in browser session.

The loader `scripts/load-tracker-cookies.sh` auto-invokes before every `boba-svc up`, `boba-svc restart`, `install.sh` Stage 6, and `start.sh` boot — so refreshing a session is a one-step *browser re-export* → *restart*. Source directory defaults to `$HOME/Downloads`, overridable via `TRACKER_COOKIE_DIR` or `--dir <path>`.

See the **Tracker Credentials Manual** for the per-tracker matrix, browser-export step-by-step, precedence rules, and the security posture (§11.4.10.A leak audit, cookie-value-never-logged). Common questions: **FAQ**.

Downloads and directories the system creates are owned by the account that started it — no chown step after every download. See the **File Ownership Guide** for why that requires `PUID=0` under rootless Podman, how to verify it on your own machine, and how to repair content created before the fix.

## At a glance

| Category              | Mandatory   | Optional                                      | Documentation      |
|-----------------------|-------------|-----------------------------------------------|--------------------|
| qBittorrent WebUI     | —           | WEBUI_USERNAME, WEBUI_PASSWORD                | <a href="#">§1</a> |
| Private trackers      | per-tracker | —                                             | <a href="#">§2</a> |
| Public tracker uplift | —           | JACKETT_API_KEY, ...                          | <a href="#">§3</a> |
| Metadata enrichment   | —           | TMDB_API_KEY, TVDB_API_KEY, ...               | <a href="#">§4</a> |
| Security scanning     | —           | SNYK_TOKEN, SONAR_TOKEN, ...                  | <a href="#">§5</a> |
| Observability         | —           | GRAFANA_USER, GRAFANA_PASSWORD                | <a href="#">§6</a> |
| Orchestrator tuning   | —           | ALLOWED_ORIGINS, MAX_CONCURRENT_TRACKERS, ... | <a href="#">§7</a> |

## Where to register (fast links)

| Provider  | Signup                                                                                                      | Notes               |
|-----------|-------------------------------------------------------------------------------------------------------------|---------------------|
| RuTracker | <a href="https://rutracker.org/forum/register.php">https://rutracker.org/forum/register.php</a>             | Username + password |
| Kinozal   | <a href="https://kinozal.tv/signup.php">https://kinozal.tv/signup.php</a>                                   | May require invite  |
| NNM-Club  | <a href="https://nnmclub.to/forum/ucp.php?mode=register">https://nnmclub.to/forum/ucp.php?mode=register</a> | Cookie-based auth   |

| Provider    | Signup                                                                                                | Notes                           |
|-------------|-------------------------------------------------------------------------------------------------------|---------------------------------|
| IPorrents   | <a href="https://iptorrents.com/">https://iptorrents.com/</a>                                         | Invite-only; freeleech enforced |
| Jackett     | <a href="https://github.com/Jackett/Jackett">https://github.com/Jackett/Jackett</a>                   | Self-hosted indexer             |
| TMDb        | <a href="https://www.themoviedb.org/signup">https://www.themoviedb.org/signup</a>                     | v3 auth API key                 |
| TVDb        | <a href="https://thetvdb.com/api-information">https://thetvdb.com/api-information</a>                 | Subscription API                |
| MusicBrainz | <a href="https://musicbrainz.org/doc/MusicBrainz_API">https://musicbrainz.org/doc/MusicBrainz_API</a> | Free — needs UA only            |
| Snyk        | <a href="https://app.snyk.io/">https://app.snyk.io/</a>                                               | Account → Auth Token            |
| SonarCloud  | <a href="https://sonarcloud.io/account/security/">https://sonarcloud.io/account/security/</a>         | Or run the compose SonarQube    |
| Gitleaks    | <a href="https://gitleaks.io/">https://gitleaks.io/</a>                                               | Optional commercial license     |

## Documentation

Everything the platform offers, indexed:

### Tracked-Items + Status Documents

The mandated entry point (§11.4.57 / §11.4.212) for every tracker and status document in this repository — Last modified and Revision are sourced verbatim from each document’s own revision header (§11.4.44); a – means the document currently carries no revision header or has no HTML/PDF export yet (never fabricated). Regenerated by `scripts/testing/update_readme_doc_links.sh` — never hand-typed (§11.4.57).

| Document       | Last modified        | Revision | Markd                 |
|----------------|----------------------|----------|-----------------------|
| Issues         | 2026-08-21T19:41:44Z | 40       | <a href="#">Markd</a> |
| Issues_Summary | —                    | —        | <a href="#">Markd</a> |
| Fixed          | 2026-08-21T19:41:44Z | 25       | <a href="#">Markd</a> |

| Document                            | Last modified        | Revision | Markd                 |
|-------------------------------------|----------------------|----------|-----------------------|
| Fixed_Summary                       | —                    | —        | <a href="#">Markd</a> |
| CONTINUATION                        | 2026-08-20T12:17:48Z | 27       | <a href="#">Markd</a> |
| PORTING-FROM-LAVA                   | 2026-07-01T16:18:43Z | 1        | <a href="#">Markd</a> |
| REMAINING_WORK_PLAN                 | 2026-08-07T19:10:54Z | 2        | <a href="#">Markd</a> |
| COMPLETION_STATUS                   | 2026-06-06T00:00:00Z | 1        | <a href="#">Markd</a> |
| RELEASE_READINESS_20260616          | 2026-06-16T10:30:00Z | 1        | <a href="#">Markd</a> |
| QA_DISCOVERY_LEDGER                 | 2026-08-21T15:05:00Z | 16       | <a href="#">Markd</a> |
| GOVERNANCE_AUDIT_2026-08-07         | 2026-08-07T19:10:54Z | 1        | <a href="#">Markd</a> |
| GOVERNANCE_AUDIT_2026-08-08_ROUND2  | 2026-08-09T12:54:30Z | 6        | <a href="#">Markd</a> |
| browser_extension/Status            | 2026-08-20T12:27:23Z | 16       | <a href="#">Markd</a> |
| browser_extension/Status_Summary    | 2026-08-20T12:27:23Z | 4        | <a href="#">Markd</a> |
| browser_extension/RELEASE_READINESS | 2026-06-13T13:10:00Z | 4        | <a href="#">Markd</a> |
| codegraph/Status                    | 2026-08-18T13:33:41Z | 2        | <a href="#">Markd</a> |
| codegraph/Status_Summary            | 2026-08-20T12:27:23Z | 1        | <a href="#">Markd</a> |
| features/Status                     | 2026-08-21T00:00:00Z | 10       | <a href="#">Markd</a> |
| features/Status_Summary             | 2026-08-20T12:27:23Z | 8        | <a href="#">Markd</a> |

## Getting started & operation

- [docs/README.md](#) — secondary index of every file directly under docs/
- [docs/USER\\_MANUAL.md](#) — end-user walkthrough
- [docs/TOKENS\\_AND\\_KEYS.md](#) — **credentials / tokens / env vars (mandatory vs optional + registration links)**
- [docs/DESIGN\\_SYSTEM.md](#) — runtime-switchable palette catalogue (8 palettes, dark + light, Darcula default)
- [docs/PLUGINS.md](#) — the 43 plugin engines
- [docs/PLUGIN\\_TROUBLESHOOTING.md](#) — what to check when a plugin breaks
- [docs/MAGNET\\_LINKS.md](#) — magnet link implementation guide
- [docs/DOWNLOAD\\_FIX.md](#) — RuTracker plugin download fix
- [docs/RELEASE\\_TORRENT\\_UPLOAD\\_FIX.md](#) — release report: torrent file upload fix
- [docs/JACKETT\\_INTEGRATION.md](#) — Jackett integration
- [docs/DEAD\\_TRACKERS\\_EXPLAINED.md](#) — dead public tracker status
- [docs/TRACKER\\_AUTH\\_TESTING.md](#) — tracker credentials live testing

- [docs/MERGE\\_SEARCH\\_DIAGNOSTICS.md](#) — merge-search diagnostics + rebuild/restart contract

## Architecture & subsystems

- [docs/architecture/](#) — 5 Mermaid diagrams (topology, search lifecycle, plugin execution, bridge, shutdown)
- [docs/DATA\\_MODEL.md](#) — Pydantic schemas & lifecycle (no relational DB)
- [docs/CONCURRENCY.md](#) — asyncio semaphore + TTL caches + retry/backoff
- [docs/OBSERVABILITY.md](#) — Prometheus + Grafana + OpenTelemetry
- [docs/PERFORMANCE.md](#) — benchmark + load + stress test layout
- [docs/api/openapi.json](#) — frozen OpenAPI spec (diffed in CI)
- [docs/API\\_Summary.md](#) — API domain summary
- [docs/Architecture\\_Summary.md](#) — architecture domain summary
- [docs/CODEGRAPH.md](#) — CodeGraph code-intelligence integration (also indexed above in Tracked-Items via codegraph/Status)
- [docs/CodeGraph\\_Summary.md](#) — CodeGraph domain summary
- [docs/MULTITRACK.md](#) — multi-track development registration (Track 11 rationale)
- [docs/CROSS\\_APP\\_THEME\\_PLAN.md](#) — cross-app theme plan

## Quality & security

- [docs/SECURITY.md](#) — threat model + credential storage
- [docs/SCANNING.md](#) — Snyk + Sonar + Semgrep + Trivy + Gitleaks + bandit + pip-audit
- [docs/QUALITY\\_STACK.md](#) — the opt-in docker-compose.quality.yml stack
- [docs/AGENT\\_GUARDRAILS.md](#) — §11.4.109 anti-forgetting enforcement (subagent constitutional preamble + orchestrator pre-action checklist)
- [docs/CONSTITUTION\\_ADDENDUM\\_QUALITY.md](#) — constitution addendum: quality stack (Principle I exception)
- [docs/PLUGIN\\_AUDIT.md](#) — plugin audit matrix
- [docs/TEST\\_RESULTS.md](#) — snapshot of last full test run
- [docs/TEST\\_SUITE\\_GUIDE.md](#) — test suite guide
- [docs/testing/](#) — DDoS resilience + test-type matrix notes

## Incidents & QA evidence

Per §11.4.238, automated HelixQA coverage is the discovery layer — manual/operator/agent-found defects are confirmation-only, and every out-of-band finding gets a coverage-escape-audit entry here.

- [docs/QA\\_DISCOVERY\\_LEDGER.md](#) — the discovery-channel ledger + coverage-escape schema (also indexed above in Tracked-Items)
- [docs/incidents/](#) — host-safety + CONST-033 forensic investigations, including the latest [2026-08-18 2nd perceived forced-logout](#) triage (root cause: partial, §11.4.6 UNCONFIRMED: boundary honestly stated; preventive gate: the resource-pressure signature challenge above)
- [docs/qa/](#) — per-item captured machine evidence (§11.4.5 / §11.4.69 / §11.4.83), one directory per workable item / task run — e.g. [docs/qa/BOB-116/](#)
- [docs/history/BOB-079-attributed-auto-commit-history.md](#) — retroactive, evidenced attribution for the 14 unattributed Auto-commit/sync: commits in v1.0.0-rc..HEAD (BOB-079/RD2-12). Attribution is read from each commit's DIFF, never its subject line; 3 of 14 are evidence-backed and 11 are recorded literally as UNKNOWN (§11.4.6 — a fabricated attribution is worse than an absent one). A forward-only record, never a history rewrite (§11.4.113)
- [docs/HOST\\_POWER\\_MANAGEMENT.md](#) — CONST-033 host power management hard ban
- [docs/guides/forced-logout-flight-recorder.md](#) — the flight-recorder guide for perceived-forced-logout triage

## Research, planning, migration & session archives

Working documents, dated research notes, superseded-implementation plans, and dated session reports — browsable indexes rather than individually curated, since new dated entries land continuously and any per-file categorization here would drift the moment a new one is added.

- [docs/research/](#) — dated investigation write-ups (tracker discovery, plugin audits, browser-extension corpus)
- [docs/superpowers/](#) — long-horizon implementation plans + design specs
- [docs/browser\\_extension/](#) — architecture, changelog, install, store listing, user guide, coverage ledger, and internal [\\_analysis/\\_plan](#) working notes (Status docs indexed above in Tracked-Items)
- [docs/migration/](#) — Python-to-Go migration spec + parity gaps

- [docs/plans/](#) — dated implementation plans
- [docs/proposals/](#) — architectural proposals under consideration
- [docs/demos/](#) — dated demo write-ups
- [docs/issues/](#) — frozen post-mortems for specific incidents (distinct from the live [docs/Issues.md](#) tracker above)
- [docs/systemd/UNITS.md](#) — systemd unit reference
- [docs/SESSION\\_REPORT\\_2026-04-19.md](#) — rolling session report
- [docs/SYSTEMATIC\\_DEBUGGING\\_2026-04-19.md](#) — systematic debugging session log
- [docs/Demos\\_Summary.md](#) — demos domain summary
- [docs/Migration\\_Summary.md](#) — migration domain summary
- [docs/Scripts\\_Summary.md](#) — scripts domain summary
- [docs/Superpowers\\_Summary.md](#) — superpowers domain summary

## Manual QA

Per §11.4.185, automated gates are necessary but not sufficient — every release deliverable requires a final human confirmation pass. Per-session manual-QA checklists live under [docs/manual-qa/](#), one file per session, in the §11.4.153 per-feature Status/video-confirmation format.

- [docs/manual-qa/2026-08-18-session-manual-qa-checklist.md](#) — 12-feature checklist covering the boba-stack reachability, cross-tracker search/dedup, download flow, BOB-112 /healthz DDoS mitigation, constitution v68 landing, resource-pressure timer, CONST-033 challenge, and pre-build gate for the 457cca4→63f3f88 session range

## Testing

- [docs/TESTING.md](#) — catalogue of every test type (30 rows)
- [docs/COVERAGE\\_BASELINE.md](#) — per-module coverage gates
- [tests/README.md](#) — layout of the tests/ tree

## Development

- [CONTRIBUTING.md](#) — contribution workflow
- [CLAUDE.md](#) — Claude Code agent protocol (TDD + rebuild-reboot)
- [AGENTS.md](#) — runtime development guidance
- [.specify/memory/constitution.md](#) — **binding architectural contract (v1.1.0)**
- [docs/scripts/](#) — per-script user guides (§11.4.18), one <name>.md per scripts/\*.sh / challenges/scripts/\*.sh — e.g. [commit-push-all.md](#), [capture-workable-items-db-delta.md](#), [regenerate-](#)

continuation-exports.md, check\_cm\_lan\_routes\_authenticated.md  
(BOB-228)

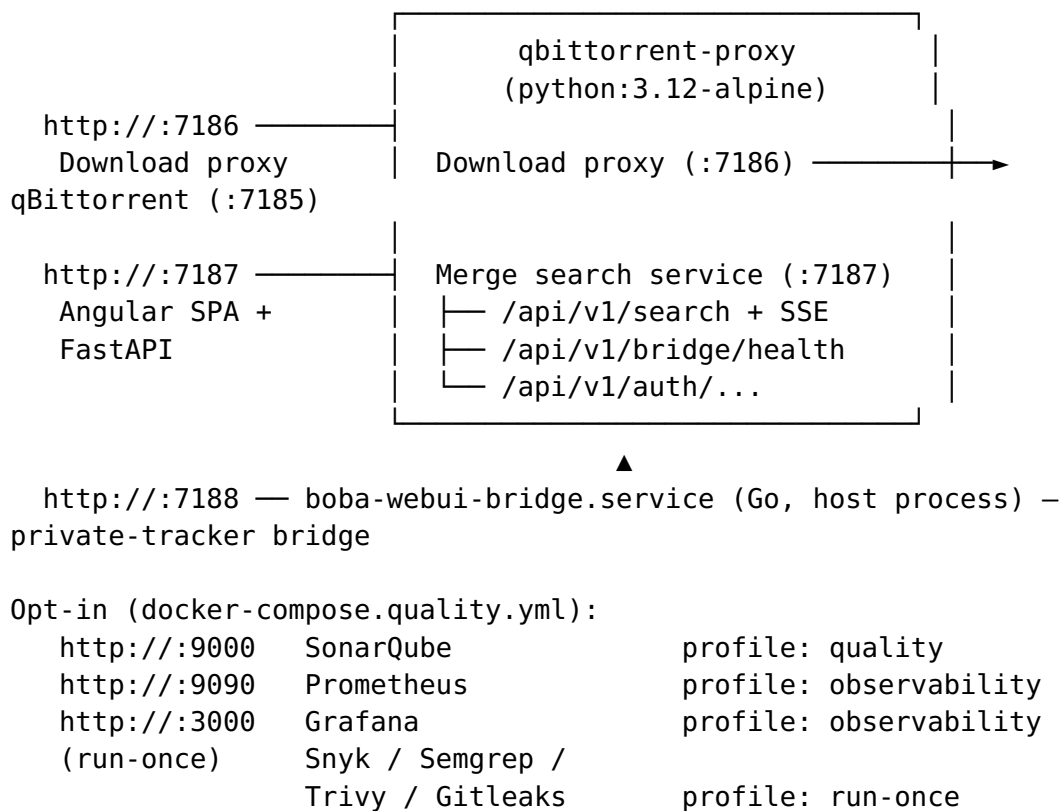
## Courses (self-paced, Asciiinema)

- courses/01-operator/ — Your first search
- courses/02-plugin-author/ — Authoring a nova3 plugin
- courses/03-contributor/ — TDD + rebuild-reboot deep dive
- courses/04-security-ops/ — Threat model + scanner bundle

## Release / CI

- CHANGELOG.md
  - releases/README.md — release artefacts layout
  - docs/OUT\_OF\_SANDBOX.md — items requiring external credentials (HelixQA / OpenCode / submodule orgs)
- 

## Architecture



Full Mermaid renders in docs/architecture/.

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# Testing

```
# Non-live-HTTP suites – run anywhere, ~10s
python3 -m pytest tests/unit/ tests/e2e/ tests/contract/ --no-cov -q

# Benchmarks – dedupe perf + live-HTTP fan-out benchmarks, ~4min
python3 -m pytest tests/benchmark/ --no-cov -q

# Security – hits the live merge service, needs stack up
python3 -m pytest tests/security/ --no-cov -q

# Integration – same
python3 -m pytest tests/integration/ --no-cov -q -m "not
    requires_credentials"

# Frontend
npx --prefix frontend ng test --watch=false

# Full scanner sweep (non-interactive; skips scanners with missing
    tokens)
./scripts/scan.sh --all

# Pre-build invariant sweep – 25 checks (workable-items integrity,
    docs-chain
# sync, resource-pressure signature, mutation-residue, §11.4.238
    discovery
# ledger freshness, ...) run before every build
./scripts/pre_build_verification.sh

# HelixQA Challenges – 31 scripts under challenges/scripts/, e.g.:
./challenges/scripts/resource_pressure_signature_challenge.sh
./challenges/scripts/ddos_resilience_challenge.sh --healthz
```

See [docs/TESTING.md](#) for the 30-row test-type catalogue.

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## Releases

Non-interactive builder at [scripts/build-releases.sh](#) produces artefacts under [releases/<version>/](#):

```
./scripts/build-releases.sh                # all targets, all
    channels
./scripts/build-releases.sh frontend        # one target
./scripts/build-releases.sh --channel release
```

Each artefact ships with SHA256SUMS + BUILD\_INFO.json.

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# Contributing

See [CONTRIBUTING.md](#) and the TDD protocol in [CLAUDE.md](#). PRs must keep the following green:

- Python unit + e2e + contract (pytest — 5077 tests collected, see docs/TESTING.md)
  - Frontend Vitest (ng test — 2743 tests collected, see docs/TESTING.md)
  - Ruff + bandit + shellcheck (via scripts/scan.sh)
- 

## License

Apache 2.0 — see [LICENSE](#).